

THINGS TO KNOW:

1. Lab report must contain following sections: (order must be maintained)
 - a) Title /Question
 - b) Theory: The brief overview of the concept /techniques/syntax/idea/tools etc. used
 - c) Steps and Commands: The steps performed along with involved commands [must be handwritten]
 - d) Output: Screenshot of the output [must be printed]
2. Output screen should be captured (use snipping tool), printed and attached in the report. [must be readable in hardcopy]
3. The output screen should reflect the unique identity of each student for that include the following things in commands (to print message in screen) and/or names and/ or paths (using stored procedure):
 - Personal Information (Name, Roll No. and Section)
4. Contents should be written on single side of A4 sized paper.
5. Cover page and contents page should be attached in the report appropriately.
6. The works must be submitted within specified deadline.

Content Page Format

Subject: Database Administration

List of Lab Works

Lab No.	Title /Question	Submission Date	Signature	Remarks
1(a)	This is sample title	2079/03/15		
1(b)	This is another title	2079/03/17		

Lab works / DBA/ BIT-6th semester

First Create PL/SQL stored procedure that displays the information about lab no. and your personal information in required places (CDB and PDB) and execute the stored procedure at beginning or end of the each lab task (for those lab works where SQL Plus is used)

Sample PL/SQL stored procedure:

```
CREATE OR REPLACE PROCEDURE my_information IS
BEGIN
    DBMS_OUTPUT.PUT_LINE('Name: Ram Prasad Sharma/Roll No.: 102 / Section: A');
END my_information;
/
```

Execution:

```
SET SERVEROUTPUT ON;
EXEC my_information;
```

1. Exploring the Multitenant Architecture of Oracle 19c.

[Hints: Execute the commands in SQL Plus to show current connection name, information about CDBs and PDBs with information about 'open_mode' and switching from CDB to PDB, changing 'open_mode' etc. Include corresponding theories, steps and outputs in report.]

2. Working with startup and shutdown modes in Oracle.

[Hints: Execute the commands for different startup modes (like nomount, mount, open, force, startup, suspend etc.) and different shutdown modes (like normal, transactional, immediate, abort etc.). Include corresponding theories, steps and outputs in report.]

3. Using Oracle Net Configuration Assistant (NetCA) for Net Service Configuration.

[Hints: Explain the idea about listener, TNS names, listeners.ora file, tnsname.ora file, NetCA, Perform the steps for adding, testing, deleting listener and Local Net Service Name configuration using NetCA. Also execute the commands for starting/ stopping listeners and reloading listener control. Include corresponding theories, steps and outputs in report]

4. Unlocking HR-Schema in Oracle 19c and connecting with hr user.

[Hints: Create A TNS service (if not created earlier), Switch the Container [to ORCLPDB]
And Unlock the Account by using the command 'ALTER USER hr IDENTIFIED BY
<your_pass> ACCOUNT UNLOCK' now connect to the system with hr user. Include
corresponding theories, steps and outputs in report]

5. Displaying basic information in oracle database and Formatting Output in SQL Plus

Execute required commands to perform following activities

- Viewing user, account_ status and default Tablespaces
- Displaying Current User in the system
- Displaying Privilege granted to a user
- Displaying all the users in the database
- Formatting output in the SQL Plus: Setting line size, formatting column and
limiting number of results

6. Working with tablespace and data file

a) Displaying tablespaces

[Hints: In theory, write about DBA_TABLESPACES view, USER_TABLESPACES
view and V\$TABLESPACE view]

- Use **DBA_TABLESPACES** view to display information about all tablespaces in
the database
 - Login to the database as **hr** user and display the **tablespace_name** and **status**
associated with current user using USER_TABLESPACES.
 - Display real-time information about the current state of tablespaces in the
database using **V\$TABLESPACE** View
 - List the tablespaces (names) in database with their types (like temporary,
permanent, undo).
- b) Create a permanent tablespace and display that the newly created tablespace in the list
of tablespaces. (The name of tablespace should be *yourName_tbs1_yourRollNo* and
data file name should be *yourName_dfl_yourRollNo*. Example: if your name is ram
and roll no. is 102 then tablespace name should be *ram_tbs1_102* and data file name
should be *ram_dfl_102*)
- c) Modify the tablespace created above as follows:
- *First Change the status to online/ offline*
 - *Now Add a new datafile to the tablespace*

— Finally rename the tablespace including your surname in tablespace name.

- d) Create a tablespace and display that the newly created tablespace in the list of tablespaces (the table space name should be test_tbs_by_YourName). Now Delete this tablespace and again show the list of tablespaces in database where this tablespace will not be listed.)
 - e) Displaying data files
[Hints: In theory, write about DBA_DATA_FILES view, V\$DATAFILE view]
— Display the information about datafiles in Database using the DBA_DATA_FILES view
— Display the information about datafiles in database V\$DATAFILE views.
 - f) Resize the data file created in (b).
- 7. Working with Oracle SQL Developer**

Install Oracle SQL Developer in your system and create two database connections one for system schema and one for hr schema. Now connect with both and execute some commands from SQL Worksheets.

[Hints: In theory, write about Sql Developer. List the steps with screenshots to create database connection in along with necessary commands in 'steps and commands'. Put the screenshots of worksheet and query result.]